Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A white precipitate is obtained on hydrolysis of

(A) PCl_5

(B) NCl_3

(C) $BiCl_3$

(D) $AsCl_3$

Q.2

The correct order of acidic strength is

(A) $C_2O_7 > SO_3 > P_4O_{10}$

(B) $CO_2 > N_2O_4 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$

(D) $K_2O > CaO > MgO$

Q.3

The major compound from which iodine is isolated is

(A) NaI

(B) $NaIO_3$

(C) $NalO_4$

(D) I_2O_s

Q.4

The most powexful oxidant among the following is

(A) HOCl

(B) $HClO_2$

(C) $HClO_3$

(D) $HClO_4$

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Q.5

Which reaction is not valid

(A) $HCl + F_2 \rightarrow HF + Cl_2$

(B) $\mathrm{HF} + \mathrm{Cl}_2 \rightarrow F_2 + HCl$

(C) $Zn + HCl \rightarrow ZnCl_2 + H_2$

(D) $Al + HCl \rightarrow AlC_3 + H_2$

Q.6

Mark the element which shows only one oxdation state

(A) F

(B) Cl

(C) Br

(D) I

Q.7

The correct order of the thermal stability of hydrogen halides $(\mathrm{H}-\mathrm{X})$ is

(A) $HCl > HBr > HI > HF$

(B) $\mathrm{HF} > \mathrm{HCl} > \mathrm{HBr} > \mathrm{HI}$

(C) $\mathrm{HCl} < \mathrm{HI} > \mathrm{HBr} > \mathrm{HF}$

(D) $HCl > HF > HBr > HI$

Q.8

Which of the following is not correctly matched?

(A) SF_4 – gas

(B) SeF_4 – liquid

(C) TeF_4 – solid

(D) SF_6 – solid

Q.9

The shape of the molecule $\mathrm{SF}_3\mathrm{Cl}_3$ is

(A) trigonal bipyramidal

(B) cubic

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(C) octahedral

(D) tetrahedral

Q.10

What are the products formed when chlorine is passed through aqueous hypo solution?

(A) $Na_2SO_3 + HCl + S$

(B) $Na_2SO_3 + SO_3 + HCl$

(C) $Na_2SO_4 + HCl + S$

(D) $Na_2SO_4 + HCl + SO_2$

Q.11

Excess of KI reacts with $CuSO_4$ solution and then $Na_2\ S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction ?

(A) $Na_2\ S_2O_3$ is oxidised

(B) CuI_2 is formed

(C) Cu_2I_2 is formed

(D) Evolved I_2 is reduced

Q.12

Of the interhalogen AX_3 compounds, ClF_3 is most reactive but BrF_3 has higher conductance in liquid state. This is because

(A) BrF_3 has higher molecular mass.

(B) ClF_3 is more volatile.

(C) BrF_3 dissociates into BrF_2^+ and BrF_4^- most easily.

(D) Electrical conductance does not depend on concentration.

Q.13

Which of the following is the best description for the behaviour of bromine in the reaction given below ?

$H_2O + Br_2 \rightarrow HOBr + HBr$

(A) Proton acceptor only

(B) Both oxidized and reduced

(C) Oxidized only

(D) Reduced only

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Q.14

Oxidation of thiosulphate with iodine gives

(A) sulphate ion

(B) sulphite ion

(C) tetrathionate ion

(D) sulphide ion

Q.15

Which of the following is correct about the reaction?

$$3NaClO \xrightarrow{heat} NaClO_3 + 2NaCl$$

(A) It is disproportionation reaction

(B) Oxidation number of Cl decreases as well as increases in this reaction

(C) This reaction is used for the manufacture of halates

(D) All of these

Q.16

The reaction of $KMnO_4$ and HCl results in

(A) oxidation of Mn in $KMnO_4$ and production of Cl_2.

(B) reduction of Mn in $KMnO_4$ and production of H_2

(C) oxidation of Mn in $KMnO_4$ and production of H_2

(D) reduction of Mn in $KMnO_4$ and production of Cl_2.

Q.17

The correct order of reactivity of halogens with alkalies is

(A) $F > Cl > Br > I$

(B) $F < Cl > Br < I$

(C) $F < Cl < Br < I$

(D) $F < Cl < Br > I$

Q.18

Among the following oxoacids, the correct order of acid strength is

(A) $HClO_2 > HClO_4 > HClO_3 > HOCl$

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(B) $HOCl > HClO_2 > HClO_3 > HClO_4$

(C) $HClO_4 > HOCl > HClO_2 > HClO_3$

(D) $HClO_4 > HClO_3 > HClO_2 > HClO$

Q.19

Among the oxides, Mn_2O_7 (I), V_2O_3 (II), V_2O_5 (III), CrO (IV, and Cr_2O_3 (V), the basic oxides are

(A) I and II

(B) II and III

(C) III and IV

(D) II and IV

Q.20

Which of the following species is not a pseudo halide :-

(A) CNO^-

(B) $RCOO^-$

(C) CN^-

(D) N_3^-

Answers & Solutions

Q.1 Answer:

$BiCl_3$

Solution:

$BiCl_3 + H_2O \rightarrow BiOCl + 2HCl \cdot BiOCl$

gives white ppt. which is used as white pigment, under the name of pearl white.

Q.2 Answer:

$C_2O_7 > SO_3 > P_4O_{10}$

Solution:

Acióic character of oxides increases as we move from left to right in a periodic table.

Q.3 Answer:

$NaIO_3$

Solution:

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Iodine is mostly isolated from sodium iodate.

Q.4 Answer:

HOCl

Solution:

The oxidizing power order is $\mathrm{HClO} > \mathrm{HClO_2} > \mathrm{HClO_3} > \mathrm{HClO_4}$

Hypochlorite $\mathrm{ClO^-}$

Chlorite $\mathrm{ClO_2^-}$

Chlorate $\mathrm{ClO_3^-}$

Perchlorate $\mathrm{ClO_4^{+7}}$

more the oxidation state less the oxidising power

The oxidation state is least in $\mathrm{ClO^-}$

$\therefore$ it is the most powerful oxidant.

Q.5 Answer:

$\mathrm{HF} + \mathrm{Cl_2} \rightarrow F_2 + HCl$

Solution:

Fluorine is more electronegative than chlorine heace, it can displace Cl **from** HCl **while chlorine cannot displace fluorine from** HF**. Therefore the following reaction is not valid.**

$HF + Cl_2 \rightarrow F_2 + HCl$

Q.6 Answer:

F

Solution:

Fluorine always shows only one oxidation state. Its oxidation state in all its compounds is -1**.**

Q.7 Answer:

$\mathrm{HF} > \mathrm{HCl} > \mathrm{HBr} > \mathrm{HI}$

Solution:

$\mathrm{HF} > \mathrm{HCl} > \mathrm{HBr} >$ **HI (Thermal stability).**

Q.8 Answer:

$\mathrm{SF_6}-$ solid

Solution:

All hexafluorides of group 16 elements are gaseous in nature.

Q.9 Answer:

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octahedral

Solution:

Hybridisation is $sp^3\ d^2$, hence octahedral

Q.10 Answer:

$Na_2SO_4 + HCl + S$

Solution:

$Na_2\ S_2O_3 + Cl_2 + H_2O \longrightarrow$
$2HCl + S + Na_2SO_4$

Q.11 Answer:

CuI_2 is formed

Solution:

$$\overset{-1}{4KI} + 2CuSO_4 \rightarrow \overset{0}{I_2} + Cu_2I_2 + 2K_2SO_4$$

$$\overset{0}{I_2} + 2\overset{2+}{Na_2}S_2O_3 \rightarrow \overset{-2.5}{Na_2}S_4O_6 + 2\overset{-1}{NaI}$$

In this CuI_2 is not formed.

Q.12 Answer:

BrF_3 dissociates into BrF_2^+ and BrF_4^- most easily.

Solution:

In liquid state BrF_3 dissociates into BrF_2 and BrF_4^- ions most easily.

Q.13 Answer:

Both oxidized and reduced

Solution:

$$H_2O + \overset{0}{Br_2} \longrightarrow \overset{+1}{HO}Br + \overset{-1}{H}Br$$

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Thus here oxidation number of Br increases from 0 to $+1$ and also decreases from 0 to -1. Thus it is oxidised as well as reduced.

Q.14 Answer:

tetrathionate ion

Solution:

$$2Na_2S_2O_3 + I_2 \rightarrow \underset{\text{Sodium tetrathionate}}{Na_2S_4O_6} + 2NaI$$

Q.15 Answer:

All of these

Solution:

$$\overset{+1}{NaClO} \rightarrow \overset{+5}{NaClO_3} + 2\overset{-1}{NaCl}.$$

All statements are correct as evident from the reaction.

Q.16 Answer:

reduction of Mn in $KMnO_4$ and production of Cl_2.

Solution:

$$2\overset{+7}{KMnO_4} + 16HCl \rightarrow 2\overset{+2}{MnCl_2} + 2KCl + 8H_2O + 5Cl_2$$

O.S of Mn changes from $+7$ to $+2$ hence reduction occurs and Cl_2 is formed.

Q.17 Answer:

$F > Cl > Br > I$

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Solution:

Reactivity follows the order $F > Cl > Br > I$

Q.18 Answer:

$HClO_4 > HClO_3 > HClO_2 > HClO$

Solution:

The larger the number of oxygen atoms attached to chlorine, greater the electron pull towards oxygen, hence, more easy to remove hydrogen from the acids. The given acids are

$$\underset{(HOCl)}{Cl-OH;} \quad \underset{(HClO_2)}{OCl-OH;} \quad \underset{(HClO_3)}{O_2Cl-OH,} \quad \underset{(HClO_4)}{O_3Cl-OH}$$

Q.19 Answer:

II and IV

Solution:

....

Q.20 Answer:

$RCOO^-$

Solution:

In pseudo halide 'N' must be present.

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